

NAPCO Nucleus

An AI agent platform for MVP Access engineering

Built by Mohammad Kamrul Hasan · AI-Augmented QA Architect · April 2026

WHAT IT IS

An AI agent that reads our client emails and meeting recordings every two hours, files them as tasks in GitLab, runs our test suites every night, classifies the failures, and ships two consolidated emails every morning. A short executive summary for leadership and a detailed test report for the team. Real production deployment, not a prototype.

9	31	2	2	\$200
WORKFLOWS	MCP TOOLS	DIMENSIONS	DAILY EMAILS	PER MONTH

TWO OPERATIONAL DIMENSIONS

- 1. Project Management.** Pulls client requirements out of email and Drive recordings every two hours. Splits each one into 3-hour tasks. Files them in GitLab with two-layer dedup so re-runs never duplicate.
- 2. Test Automation.** Runs API Functional, API Integration, API Load, and MVP Access E2E suites on schedule. Classifies failures as real bug, regression, flaky, environment, or known issue. Composes one Daily Report PDF every morning at 09:00 BDT.

HOW ONE RUN WORKS

- 1 Trigger fires**
GitHub Actions cron or workflow_dispatch starts the run on the self-hosted Windows runner.
- 2 agent.py boots**
Loads .env with override semantics, builds the MCP server, loads the system prompt plus the task prompt.
- 3 Claude reasons**
The Claude Agent SDK opens a session against the locally-installed Claude Code CLI. No API key. Cost is fixed monthly.
- 4 Tools run**
Claude calls MCP tools as needed. Each tool wraps one external system: IMAP, Drive, GitLab, Newman, pytest, Playwright, Locust, SMTP.
- 5 Memory persists**
Activity log, requirements seen, test history all written to nucleus_memory.db (SQLite plus FTS5). The DB is committed to the repo.
- 6 Process exits**
One Claude turn per process. Clean shutdown. The next workflow starts with full memory of every prior run.

THE EIGHT WORKFLOWS

Each one runs on its own schedule and writes to memory. The Daily Report reads memory and composes one consolidated picture every morning.

#	Workflow	Schedule	What it does
1	API Functional Test	02:00 BDT daily	Newman and Postman collection across the API surface.
2	API Integration Test	02:00 BDT daily	pytest integration suite with regression diff against prior runs.
3	API Load Test	02:00 BDT daily	Locust multi-tier from 10 to 10,000 users with server-recovery cooldowns.
4	MVP Access E2E Test	02:00 BDT daily	Playwright full suite. Failure screenshots embedded in the PDF.
5	Daily Report (Detailed)	09:00 BDT daily	Composes the 4-test detailed PDF and emails it to the FULL TEAM.
6	Daily Report (Summary)	09:30 BDT daily	6-block executive summary (4 tests + CICD + Runner) to LEADERSHIP.
7	Requirement Management	Every 2h business hours	IMAP plus Drive ingest. Splits requirements into 3h tasks. Files in GitLab.
8	MVPAccess CICD	22:00 BDT daily	TFS pull, MSBuild Release, IIS deploy via UNC, health check, memory log.
9	Probe Runner Filesystem	Manual	Diagnostic. Inspects runner state during triage.

THE 31 MCP TOOLS

Every tool wraps one external system. Tools do not contain logic. Reasoning lives in the prompts.

Submodule	N	What it does
memory	5	Recall activity logs, search requirements, recall test runs, write memory rows.
requirements	4	Poll IMAP, ingest Drive recordings, read inbox, publish to GitLab.
tests	9	Run Newman, pytest, Locust tiers, Playwright suites, single specs, health probes.
files	5	List, read, write, edit files in sibling projects. Capture Playwright a11y snapshots.
git	3	Diff, commit and push, recent commits across any project.
report	5	Generate PDF, send email, post Teams card, tail logs, clean reports folder.

ARCHITECTURAL PRINCIPLES

- **Single source of truth for secrets.** One .env at the project root, gitignored. The agent never reaches into another project for credentials.
- **No ANTHROPIC_API_KEY anywhere.** Reasoning runs through the local Claude Code CLI under a Claude Max subscription. Cost is fixed monthly, not per token.
- **Memory committed to the repo.** nucleus_memory.db lives in git. Cloning the project on a new machine instantly recovers prior context and dedup state.
- **Algorithms in prompts, not Python.** Tools wrap I/O only. Classification, summarization, and regression analysis happen in markdown that Claude executes.
- **One Claude turn per process.** agent.py loads, runs ONE turn, exits. Clean memory, no leaked state, easy to reason about.
- **Two consolidated emails per day.** At 09:00 BDT the detailed test report goes to the full team. At 09:30 BDT a 7-block executive summary goes to leadership.

TECH STACK

Reasoning: Claude Agent SDK, Claude Code CLI, Claude Max subscription. **Runtime:** Python 3.13, GitHub Actions on a self-hosted Windows VM runner. **Memory:** SQLite with FTS5 fuzzy search. **Reporting:** Reportlab. **Tests it orchestrates:** Newman, pytest, Locust, Playwright. **External integrations:** IMAP and SMTP (Gmail), Google Drive API, Groq Whisper, GitLab REST v4, Microsoft Teams webhook.

WHAT THIS DELIVERS

FOR THE TEAM	FOR LEADERSHIP	FOR THE QA ROLE
Client requirements arrive in GitLab within two hours. Test results land in one morning email instead of six fragments scattered through the night.	One source of truth for what changed, what tested, what regressed, what shipped, what the client asked for. Every morning. Nothing fabricated.	A working demonstration that one architect who knows how to direct AI ships production infrastructure that previously required a senior dev team.

THE THESIS

QA architect plus AI equals senior developer team output. NAPCO Nucleus is the proof.